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GENETIC DIVERSITY AND EVOLUTION

Convergent Co-option of the Retroviral *gag* Gene during the Early Evolution of Mammals

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ABSTRACT Endogenous retroviruses, records of past retroviral infections, are ubiquitous in vertebrate genomes. On occasion, vertebrate hosts have co-opted retroviral genes for their own biological functions. Here, we perform a phylogenomic survey of retroviral *gag* gene homologs within vertebrate genomes and identify two ancient co-opted retroviral *gag* genes, designated *wucaishi1* (*wcs1*) and *wucaishi2* (*wcs2*), in mammals. Conserved synteny and evolutionary analyses suggest that the *wcs1* and *wcs2* co-options occurred before the origin of modern placental mammals (~100 million years ago) and before the origin of modern marsupials (~80 million years ago), respectively. We found that the *wcs* genes were lost or pseudogenized multiple times during the evolutionary course of mammals. While the *wcs1* gene is mainly subject to negative selection in placental mammals (except in Perissodactyla), the *wcs2* gene underwent positive selection in marsupials. Moreover, analyses of transcriptome-sequencing (RNA-seq) data suggest that the *wcs1* and the *wcs2* genes are expressed in a wide range of tissues. The convergent *wcs* co-option in mammals implies the retroviral *gag* gene might have been repurposed more frequently than previously thought.

IMPORTANCE Retroviruses occasionally can infect host germ lines, forming endogenous retroviruses. Vertebrates, in turn, recruited retroviral genes for their own biological functions, a process formally known as co-option or exaptation. To date, co-opted retroviral *gag* genes have rarely been reported. In this study, we identified two co-opted retroviral *gag* genes, designated *wucaishi1* (*wcs1*) and *wucaishi2* (*wcs2*), in mammals. The co-option of *wcs1* and *wcs2* occurred before the origin of modern placentals and before the origin of modern marsupials, respectively. Our study indicates that retroviral *gag* gene co-option might have occurred more frequently than previously thought during the evolutionary course of vertebrates.

KEYWORDS co-option, endogenous retrovirus, paleovirology, phylogenetics

Retroviruses infect a variety of vertebrates and cause many diseases, such as AIDS and cancers. Uniquely among RNA viruses, retroviruses replicate through reverse transcription of viral RNA into DNA and integration of newly synthesized DNA into the host chromosome. While retrovirus infection usually takes place in host somatic cells, retrovirus occasionally infects host germ line cells. The integrated retroviral copies in the germ line (known as endogenous retroviruses [ERVs]) begin to be vertically inherited as host genetic elements (1, 2). ERVs are ubiquitously present and highly abundant in vertebrate genomes (3–5); for example, ERVs make up around 8% of the human genome (6). Recent comparative genomic studies have uncovered many nonretroviral sequences endogenized in diverse eukaryotes (7–10). Endogenous viral elements (EVEs), including ERVs, record past viral infections and thus provide molecular fossils to study the origin and deep history of viruses, laying the foundation of an emerging field, paleovirology (11, 12).

Most ERVs accumulate deleterious mutations and become degraded over time (2, 5).

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On occasion, vertebrate hosts can recruit ERVs for their own biological functions, a process formally termed co-option or exaptation (13, 14). Co-opted retroviral genes mediate a variety of host biological processes, such as protecting hosts from exogenous retrovirus infection (e.g., the *Fv1* gene, a co-opted *gag* gene in rodents), regulating placenta formation (e.g., the *syncytin* genes, co-opted *env* genes in placentals), and regulating the expression of host genes by co-opting retroviral regulatory sequences (12–21). While the retroviral *env* gene has been frequently co-opted in placentals, few cases of retroviral *gag* co-option have been reported, with the *Fv1* gene the best-known example (13, 14). As a restriction factor, the *Fv1* gene blocks the replication of various retroviruses, such as murine leukemia virus, lentiviruses, and foamy viruses (22, 23). Sequence analysis has revealed similarity between Fv1 and the Gag protein of murine endogenous retrovirus L (MuERV-L) (22). Conserved synteny analysis dates the integration of the *Fv1* progenitor into the genomes of Muroidea, a superfamily of rodents, back to ~45 million years ago (MYA) (24, 25). In addition to retroviral *gag* genes, retrotransposon *gag* genes have also been found to be repurposed in mammals (20); for example, the *Arc* gene, a co-opted *gag* gene derived from Ty3/gypsy retrotransposons, mediates mRNA transfer between different nerve cells (20, 21). With the recent development of next-generation sequencing, an increasing number of vertebrate genomes have been sequenced, which provide important resources for identifying more cases of retroviral gene co-option.

As illustrated by the "Red Queen" hypothesis, the virus-host conflicts result in recurrent cycles of an arms race, where hosts evolve resistance to viral infection and viruses, in turn, develop countermeasures to evade or block host defenses (26–30). Both host and virus genes involved in the genetic conflicts have been found to be subject to positive selection (26–30). Nearly all known restriction factors, host proteins that inhibit the replication of viruses (29), exhibit signatures of positive selection, including the *Fv1* gene (23–25, 29, 31, 32).

In this study, we performed a phylogenomic analysis of retroviral *gag* gene co-option events and identified two co-opted retroviral *gag* genes in mammals. The co-option events date back to the origin of modern placentals and the origin of modern marsupials, respectively. We also analyzed the evolutionary fingerprints and expression patterns of the co-opted *gag* genes.

RESULTS

Co-option of retroviral *gag* genes in mammals. Initially, to explore the evolutionary history of the *Fv1* gene, we employed a combined similarity search and phylogenetic analysis approach to identify homologs of MuERV-L Gag protein, the closest relative of Fv1, within the vertebrate proteomes. To our surprise, we identified several proteins that share significant similarity with the Gag proteins of MuERV-L in mammals but do not share high identity with Fv1 (Fig. 1; see Table S1 in the supplemental material). We designated those Gag-like proteins Wucaishi (Wcs) proteins. In Chinese mythology, wucaishi (five-colored stone) was repurposed and used by the goddess Nüwa to patch up the sky after the pillars of heaven were broken. To further explore the relationship among Wcs, Fv1, and retroviral Gag proteins (see Table S2 in the supplemental material), we performed phylogenetic analysis and found that the newly identified Wcs proteins do not group with the Fv1 proteins. The Wcs proteins cluster into two monophyletic groups, termed Wcs1 and Wcs2, with strong support (bootstrap values of 100% for both groups) (Fig. 1A and C). The Wcs1 protein is more closely related to Fv1, MuERV-L, and human endogenous retrovirus L (HERV-L) than the Wcs2 protein. These results suggest that the *wcs1* and *wcs2* genes arose through two co-option events independently of *Fv1* co-option.

Evolutionary history of the *wcs1* locus. A similarity search based on vertebrate proteomes and phylogenetic analysis suggested that the Wcs1 protein is present within at least five mammalian orders, Carnivora, Perissodactyla, Cetartiodactyla, Chiroptera, and Rodentia (Fig. 1 and 2; see Table S5 in the supplemental material). The *wcs1* gene is located between *Acid phosphatase 4* (*ACP4*) or *Acid phosphatase T* (*ACPT*) (*ACP4* and

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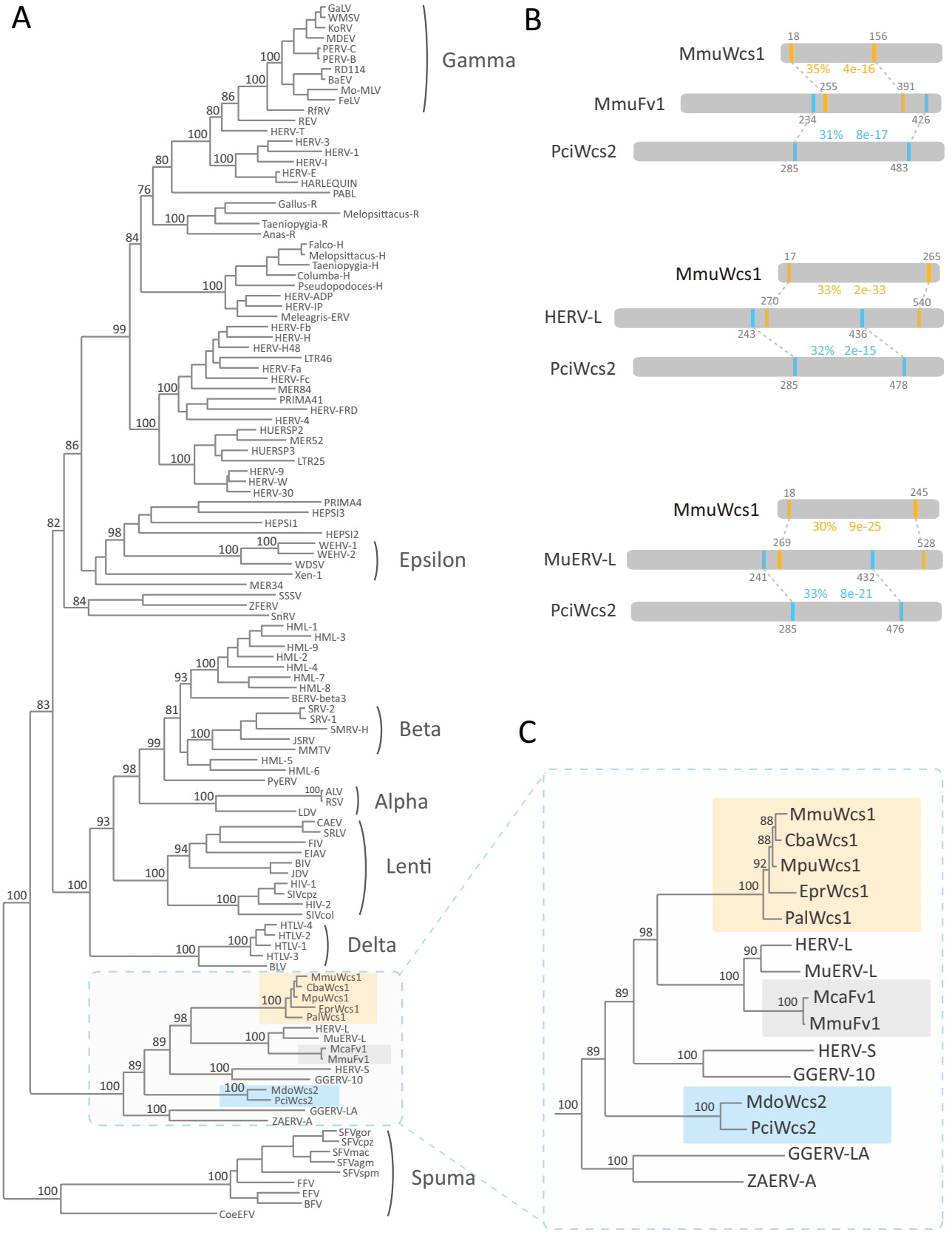


FIG 1 Relationships among the Wcs proteins, the Fv1 proteins, and retrovirus Gag proteins. (A) Phylogenetic relationships among the Wcs proteins, the Fv1 proteins, and representative retrovirus Gag proteins. The numbers near selected nodes represent bootstrap values. The Wcs1 and Wcs2 proteins are highlighted in orange and blue, respectively. (B) Similarities among the Wcs proteins, the Fv1 protein, and HERV-L and MuERV-L Gag proteins. The

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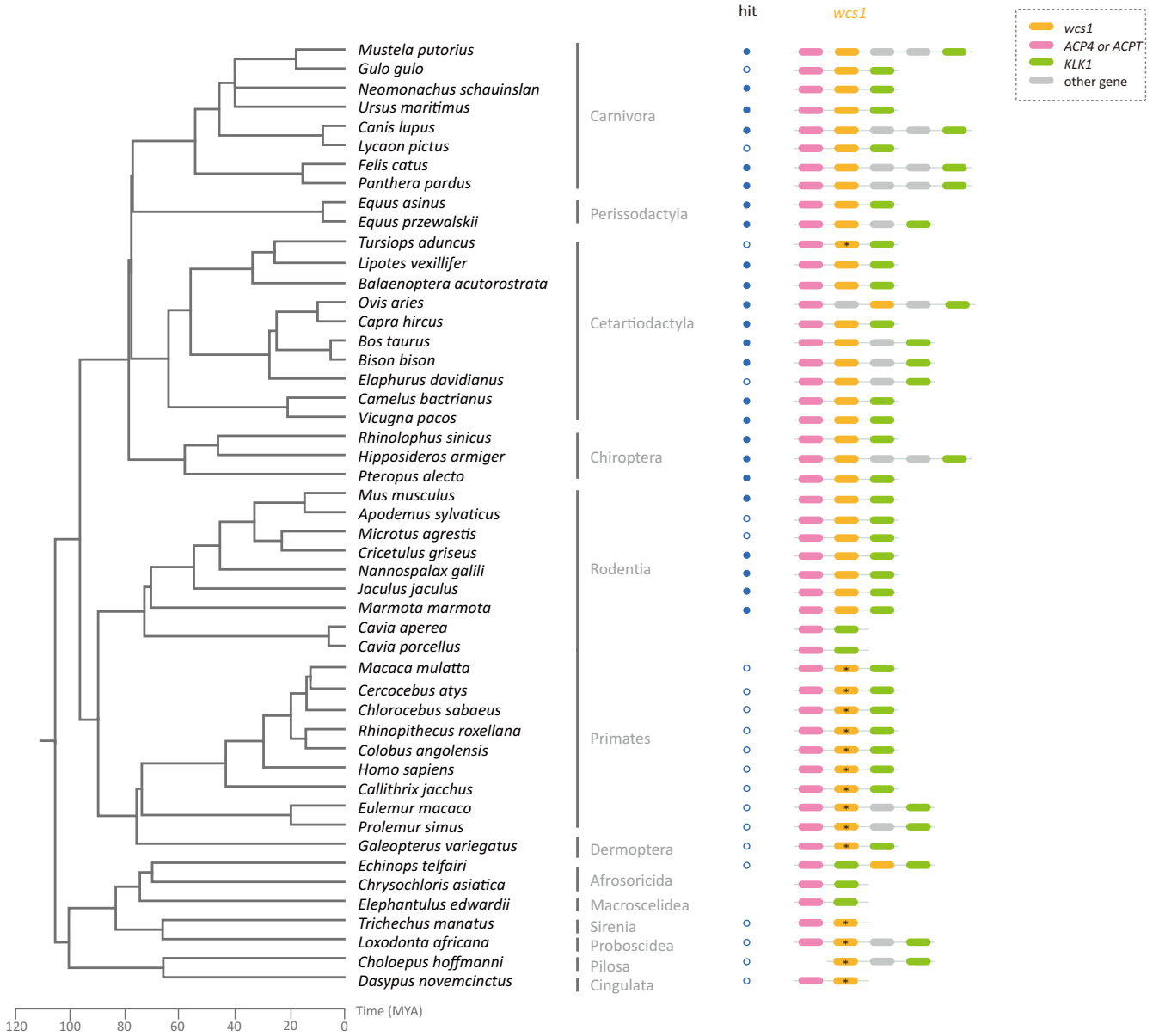


FIG 2 Conserved synteny of the *wcs1* genes. The phylogenetic relationships among placentals and the evolutionary time scale are based on TimeTree (33, 53). The solid and open circles indicate that the *wcs1* gene homologs were identified by BLASTp and BLASTn/tBLASTn, respectively. The gene synteny of *wcs1* is shown near the corresponding species. Genes with premature stop codons or frameshift mutations are labeled with asterisks.

ACPT form a monophyletic group and are essentially orthologs) (see Fig. S1 and Table S6 in the supplemental material) and *Kallikrein 1* (*KLK1*) (Fig. 2). The *ACP4- or ACPT-KLK1* synteny is conserved across placentals (Fig. 2). To further explore the evolutionary history of the *wcs1* gene, we used a combined gene synteny and similarity search approach and found sequences that share significant similarity with the *wcs1* gene and are located between *ACP4* and *KLK1* in other orders of placentals, including Primates, Dermoptera, Afrosoricida, Sirenia, and Proboscidea. Most of the *wcs1* sequences in these orders appear to be pseudogenized (Fig. 2). The *wcs1* gene appears to be lost in

FIG 1 Legend (Continued)
dashed lines indicate regions that share significant similarity between two proteins. The gray numbers indicate residue positions. The numbers in blue or orange indicate sequence identities and BLASTp E values. (C) Phylogenetic relationships among the Wcs proteins, the Fv1 protein, and HERV-L and MuERV-L Gag proteins (enlargement of the boxed area in panel A). Mca, *Mus caroli*; Mmu, *Mus musculus*; Mpu, *Mustela putorius*; Cba, *Camelus bactrianus*; Pal, *Pteropus alecto*; Epr, *Equus przewalskii*; Mdo, *Monodelphis domestica*; Pci, *Phascolarctos cinereus*.

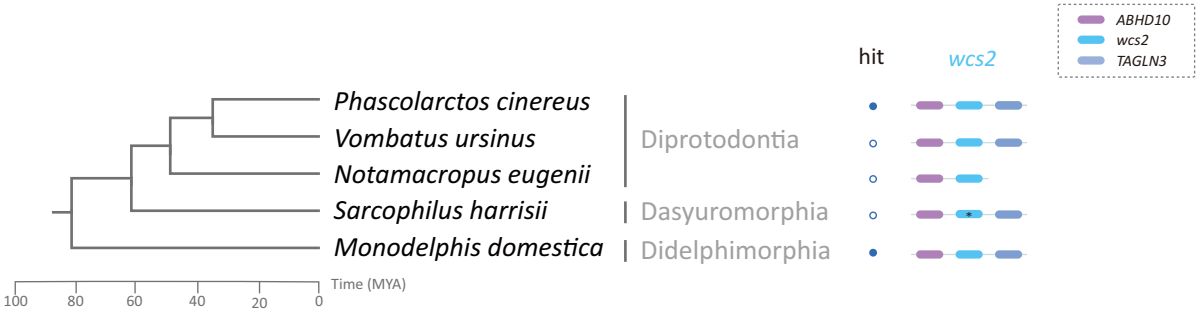


FIG 3 Conserved synteny of the *wcs2* genes. The phylogenetic relationships among marsupials and the evolutionary time scale are based on TimeTree (33, 53). The solid and open circles indicate that the *wcs2* gene homologs were identified by BLASTp and BLASTn/tBLASTn, respectively. The gene synteny of *wcs2* is shown near the corresponding species. A gene with a premature stop codon or frameshift mutation is labeled with an asterisk.

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some species of Rodentia, Afrosoricida, and Macroscelidea. The *wcs1* sequence is located only together with *KLK1* in *Choloepus hoffmanni* (order: Pilosa) and is located only together with *ACP4* in *Dasypus novemcinctus* (order: Cingulata) and *Trichechus manatus* (order: Sirenia), which might be due to the shortness of the contigs. Nevertheless, the conserved synteny of *ACP4-wcs1-KLK1* among major orders of placentals suggests that the insertion of the *wcs1* gene occurred at least in the most recent common ancestor of modern placentals, around 100 MYA (33, 53).

Evolutionary history of the *wcs2* locus. Our initial similarity search and phylogenetic analysis suggested the Wcs2 protein is present in the genomes of *Phascolarctos cinereus* and *Monodelphis domestica*. For both species, the *wcs2* gene is located between *Transgelin-3* (*TAGLN3*) and *Abhydrolase domain containing 10* (*ABHD10*) (Fig. 3). The *TAGLN3-ABHD10* synteny is conserved across vertebrates ranging from zebrafish (*Danio rerio*) to human. The different syntenies between the *wcs1* and *wcs2* loci also support the notion that the *wcs1* and the *wcs2* genes arose through independent co-option events. To further explore the evolutionary history of the *wcs2* gene, we used a combined gene synteny and similarity search approach and found sequences that share significant similarity with the *wcs2* gene and are located between *TAGLN3* and *Abdh10* in three marsupial orders, that is, Diprotodontia, Dasyuromorphia, and Didelphimorphia. In *Notamacropus eugenii*, the *wcs2* gene is located only together with *Abdh10*, which might be due to the shortness of the contig. The *wcs2* gene has been pseudogenized in some marsupial species (Fig. 3; see Table S5). Nevertheless, the conserved synteny of *TAGLN2-wcs2-ABHD10* across marsupials suggests that the co-option of *wcs2* occurred in the common ancestor of modern marsupials, around 80 MYA (33, 53).

Selection analyses of the *wcs* genes. The selection pressure that has acted on a gene can be inferred by comparing the number of nonsynonymous substitutions per nonsynonymous site (*dN*) and the number of synonymous substitutions per synonymous site (*dS*) (27–30). A *dN*/*dS* ratio of greater than one indicates positive selection. We characterized the evolutionary fingerprints in the *wcs1* gene for five placental orders and the *wcs2* gene for marsupials. First, we used the single-likelihood ancestor-counting (SLAC) method (34) to estimate the *dN*/*dS* ratios for the *wcs* genes of five placental orders and marsupials and observed generally small *dN*/*dS* values (most around 0.1), except for Perissodactyla (*dN*/*dS* = 1.95) (Table 1). The *dN*/*dS* ratios suggest the *wcs* genes might mainly undergo negative selection in mammals, except Perissodactyla. Because the *dN*/*dS* ratio appears to be a conservative statistic (35, 36), we used site models to detect positively selected sites in the *wcs* genes. The neutral model (M8a) was not rejected in favor of the model with positive selection (M8) in four placental orders. The M8a model was rejected in the *wcs* genes of Perissodactyla ($P = 0.03$) and marsupials ($P = 0.01$) (Table 1). We detected 17 sites and 9 sites under positive selection in Perissodactyla and marsupials, respectively. Moreover, we also used a fixed effects

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TABLE 1 Selection analysis of *wcs* genes in mammals

Order/class	No. of species	No. of sites	*dN*/*dS*[a]	No. of branches under positive selection/total[b]	No. (%) of sites under positive selection[c]	No. (%) of sites under negative selection[c]	2Δl[d]	*P* value[d]	No. (%) of sites under positive selection[d]	Codons with *dN*/*dS* value of >1
Carnivora	22	291	0.07	0/41		83 (28.52)	2.41	0.12		
Perissodactyla	3	259	1.95	1/3			4.79	0.03	17 (6.56)	T9[e], H20[e], V26[e], S27[e], Q31[e], L38[e], T88[e], W94[e], R97[e], L137[e], V146[e], D170[e], Q174[e], V182[e], Q186[e], T188[e], R255[e]
Cetartiodactyla	22	288	0.08	1/41		72 (25.00)	0.00	1.00		
Chiroptera	6	289	0.18	0/9		56 (19.38)	0.00	1.00		
Rodentia	25	284	0.07	1/47		128 (45.07)	0.00	0.95		
Marsupialia	4	645	0.65	0/5	1 (0.16)	34 (5.27)	6.73	0.01	9 (1.40)	L233[e], V246[e], S305[e], G336[e], K343[e], E353[e], T369[e], S389[e], Q591[e]

[a]The *dN*/*dS* values of the *wcs* genes were calculated using the SLAC method in HyPhy.

[b]Branches under positive selection with *P* values of <0.05 were detected using the aBSREL method in HyPhy.

[c]Sites under positive or negative selection with *P* values of <0.05 were detected using the FEL method in HyPhy.

[d]2Δl represents twice the difference in the natural log values of the likelihoods of the two models (M8a versus M8) being compared. The *P* value indicates the confidence with which the neutral model (M8a) can be rejected in favor of the positive-selection model (M8).

[e]Codon under positive selection with a posterior probability of >95% by Bayes empirical Bayes (BEB) analysis (54).

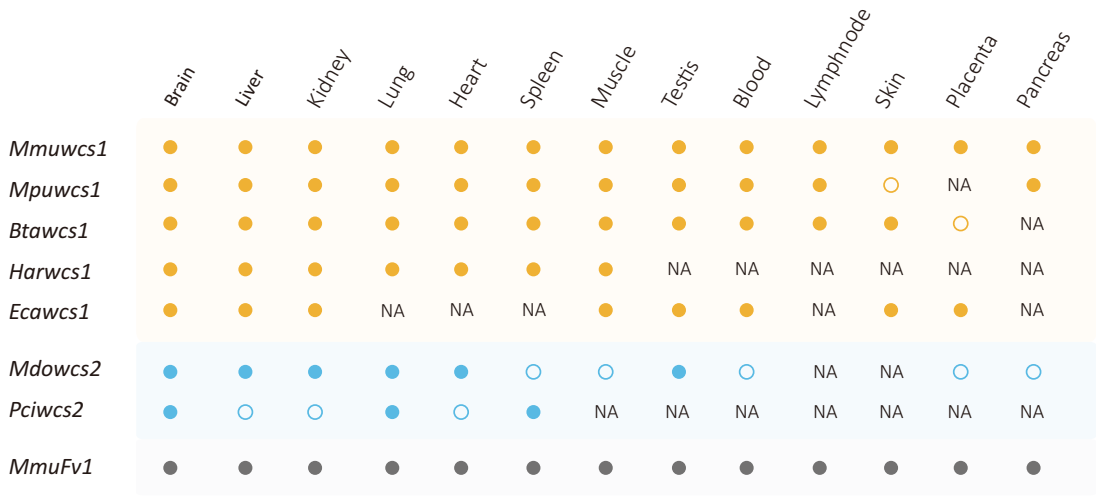


FIG 4 Expression patterns of the *wcs* genes. The solid and open circles indicate that the *wcs1* gene is expressed or is not expressed, respectively. NA indicates that there are no RNA-seq data for the organ (tissue) of the species. Mmu, *Mus musculus*; Mpu, *Mustela putorius*; Bta, *Bos Taurus*; Har, *Hipposideros armiger*; Eca, *Equus caballus*; Mdo, *Monodelphis domestica*; Pci, *Phascolarctos cinereus*.

likelihood (FEL) method to detect positively selected sites (34). For all five mammalian orders, no site of the *wcs1* gene was found to be under positive selection at a significance level of 0.05, while many sites (up to 45.07%) were found to be under negative selection (Table 1). For marsupials, there was only one site (0.16%) under positive selection but 34 sites (5.27%) under negative selection. Finally, we employed the adaptive branch site random-effects likelihood (aBSREL) method to test whether positive selection had occurred on a proportion of branches (36, 37). We found only three branches subject to positive selection for three mammalian lineages, that is, Perissodactyla (1 out of 3 branches), Cetartiodactyla (only 1 out of 41 branches), and Rodentia (only 1 out of 47 branches). Taken together, our analyses suggest that the *wcs1* gene is mainly subjected to negative selection in placentals, except Perissodactyla, and that the *wcs2* gene might undergo positive selection.

Expression patterns of the *wcs* genes. To explore the expression patterns of the *wcs* genes, we retrieved transcriptome-sequencing (RNA-seq) data for seven mammal species, that is, *Mus musculus* (order: Rodentia), *Mustela putorius* (order: Carnivora), *Bos taurus* (order: Cetartiodactyla), *Hipposideros armiger* (order: Chiroptera), *Equus caballus* (order: Perissodactyla), *Monodelphis domestica* (order: Didelphimorphia), and *Phascolarctos cinereus* (order: Diprotodontia) (Fig. 4; see Table S3 in the supplemental material). Similar to the *Fv1* gene, the *wcs1* gene was expressed in nearly all the tissues studied (Fig. 4). However, the *wcs2* gene was expressed in only some of the tissues (Fig. 4). Nevertheless, our results show that both the *wcs1* and *wcs2* genes are expressed in mammals.

DISCUSSION

Retroviral *env* genes have been frequently captured and repurposed for placentation in placental mammals and the viviparous placental *Mabuya* lizard and are known as the *syncytin* genes (17, 38, 39). Syncytins arose independently more than 10 times in placental mammals (17, 38, 39). Moreover, a captured *env* gene was found to be conserved in spiny-rayed fishes (Acanthomorpha) for more than 110 million years (40). However, unlike retroviral *env* genes, few co-opted retroviral *gag* genes have been identified to date (22, 24, 25). In this study, we identified two new co-opted retroviral *gag* genes in mammals, that is, the *wcs1* and *wcs2* genes, which arose convergently in two major linages of mammals. The *wcs1* and *wcs2* co-options occurred before the origin of modern placentals (~100 MYA) and before the origin of modern marsupials (~80 MYA), respectively (Fig. 5). Both *wcs* genes are much older than the *Fv1* gene,

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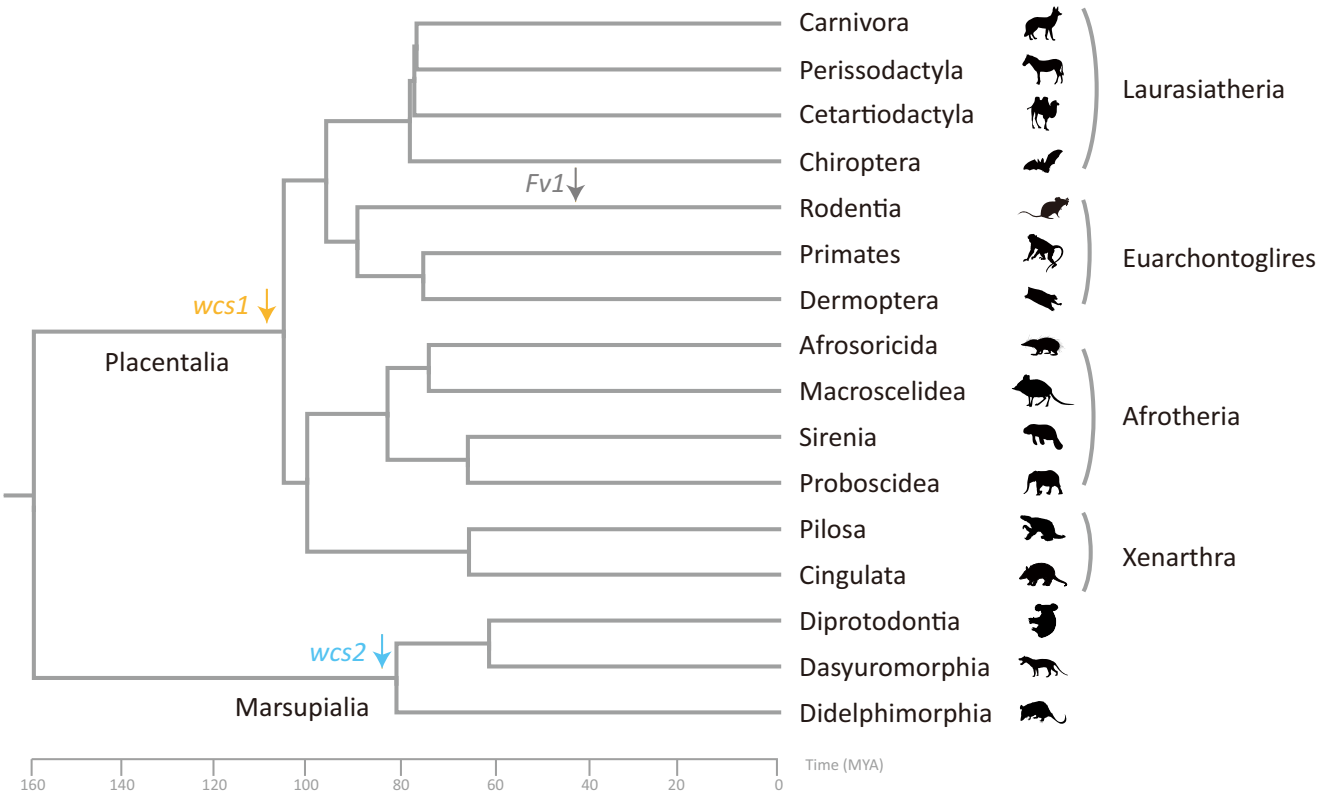


FIG 5 Evolutionary history of the *wcs* genes. The phylogenetic relationships among mammals and the evolutionary time scale are based on TimeTree (33). The *wcs1* (orange) and *wcs2* (blue) genes originated before the emergence of modern placentals (~100 MYA) and marsupials (~80 MYA), respectively. The *Fv1* gene (labeled in gray) originated in rodents ~45 MYA.

which integrated into the genomes of Muroidea ~45 MYA (24, 25). The convergent co-option during the early evolution of mammals suggests that co-option of retroviral *gag* genes might occur more frequently than previously thought.

Some of the Wcs proteins, for example, the Wcs1 protein of *M. putorius* (accession no. XP_004767240) and the Wcs1 protein of *Camelus bactrianus* (accession no. XP_010945334), were annotated as Fv1 proteins, probably because Fv1 and Wcs proteins have significant similarity to each other (Fig. 1B). However, the Wcs proteins arose independently from Fv1, and it might not be appropriate to annotate the Wcs proteins as Fv1. Therefore, caution should be exercised when annotating retroviral gene homologs in vertebrate genome-sequencing projects. We would do better to name a retroviral Gag protein homolog an "uncharacterized protein derived from retroviral Gag protein" rather than Fv1.

Synteny analysis suggests that the *wcs1* and *wcs2* genes arose once through co-opting retroviral *gag* genes in placentals and marsupials, respectively. However, the *wcs1* gene has been lost or pseudogenized multiple times during the course of placental evolution, and the *wcs2* gene was pseudogenized in Dasyuromorphia. Frequent loss or pseudogenization of the *wcs* genes in some linages, such as primates and Dasyuromorphia, suggests that the *wcs* genes might not work as essential genes in these linages. On the other hand, strong purifying selection acted on the *wcs* genes in some mammalian orders, implying that the *wcs* genes were recruited for important host functions. Moreover, the *wcs* genes are expressed in a wide range of tissues. All these lines of evidence suggest that the *wcs* genes might be functional co-opted retroviral *gag* genes in mammals.

The Fv1 protein blocks the replication of various retroviruses. Historically, the term "restriction factor" was coined following the characterization of the *Fv1* gene (29). Intuitively, it can be hypothesized that the *gag*-derived *wcs* genes might act as

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restriction factors, like the *Fv1* gene. The evolutionary arms race is expected to drive positive selection in host genes involved in host-virus conflicts (27–30). Nearly all known restriction factors exhibit strong signatures of positive selection (29). On one hand, we found some significant evidence that positive selection might have acted on the *wcs1* gene in Perissodactyla and on the *wcs2* gene in marsupials, suggesting the corresponding Wcs proteins might be involved in host-virus interactions. However, this result is based on a limited number of sequences and should be confirmed with a larger data set. On the other hand, we found no strong evidence for positive selection in the *wcs1* gene in other placental mammal lineages. The *wcs* gene seems to undergo mainly negative selection in placental mammals, suggesting that some Wcs1 proteins might mediate some biological functions other than antiviral host defense. Indeed, not all Gag-derived proteins are involved in host-virus interactions. For example, the Arc protein, which is derived from the Gag protein of a Ty3/gypsy retrotransposon (different from a retrovirus in the strict sense), mediates RNA transportation across synaptic boutons (20, 21). It is possible that the Wcs proteins mediate some biological processes in mammals other than antiviral host defense, and further work is needed to characterize the functions of the *wcs* genes.

MATERIALS AND METHODS

Identification of retroviral Gag homologs. We employed a combined similarity search and phylogenetic analysis approach to screen 261 vertebrate proteomes for homologs of the retroviral Gag proteins (see Table S4 in the supplemental material). First, we performed a similarity search against 261 vertebrate proteomes using the MuERV-L Gag protein (accession no. CAA73250.1) as the query and an E value cutoff of 10^{-5}. Next, the significant hits, Fv1, and the Gag proteins of representative retrotransposons and retroviruses were aligned using MAFFT 7 and then manually refined (41, 42). Initial phylogenetic analyses were performed using an approximate-maximum-likelihood method implemented in FastTree 2 (43). The significant hits that clustered together with retroviral Gag proteins were retrieved for synteny analysis. Among these hits, we found only two clusters of proteins, that is, Wcs1 and Wcs2, which shared conserved synteny.

Phylogenetic analyses. To further explore the phylogenetic relationships among Wcs, Fv1, and retrovirus Gag proteins, we used 7 Wcs protein sequences, 2 Fv1 protein sequences, and 109 representative retrovirus Gag protein sequences to perform a phylogenetic analysis (see Tables S1 and S2). All the protein sequences were aligned using MAFFT 7 with the L-INS-i strategy (42). The alignment was manually refined and trimmed using TrimAl 1.2b with a gt value of 0.19 to exclude ambiguous regions (44). A phylogenetic tree was inferred using a maximum-likelihood method implemented in IQ-TREE 1.6.0 with an LG-plus-F-plus-G4 amino acid substitution model (45). The best model was chosen using ProtTest 3.4 (46). The branch supports were assessed using ultrafast bootstrap with 1,000 replications (47).

Distribution and synteny of *wcs* in mammals. To further explore the distribution and synteny of the *wcs* genes in mammals, we used the BLASTn or tBLASTn algorithm with an E value cutoff of 10^{-5} to identify the *wcs* gene homologs in representative species covering a broad diversity of mammals (Fig. 2 and 3). The synteny of the *wcs* genes was identified based on gene annotation and a similarity search.

Selection analyses. To characterize the selection pressure on the *wcs* genes, we choose five mammalian orders and marsupials, including 78 *wcs1* genes and 4 *wcs2* genes without premature stop codons or frameshift mutations. We performed selection pressure analysis for each mammal order and marsupials. All the *wcs* sequences were aligned using MAFFT 7 and then manually refined (42). The gene trees of each order were reconstructed using a maximum-likelihood method implemented in PhyML 3.1 (48). The best-fit substitution models were chosen using jModelTest 2.1.10 (48, 49). The *wcs* gene trees are generally similar to the species trees. First, we used the SLAC method in the HyPhy package (34, 50) to calculate the *dN*/*dS* value for the *wcs* genes. Next, we used codeml in PAML 4.9 (51) and FEL in the HyPhy package (34, 50) to detect the codons under positive selection. For the PAML analyses, likelihood ratio tests were performed to compare two pairs of site-specific models (a neutral model versus a positive-selection model): M8a versus M8. χ^2 analyses were performed using R. For the FEL analyses, we used a *P* value of 0.05 as the cutoff value to summarize the number of sites under positive selection or negative selection. Finally, we employed the aBSREL method in the HyPhy package to detect branches under positive selection (36, 37, 50).

Expression pattern of the *wcs* genes. RNA-seq raw read sequences from 13 tissues of seven species were retrieved for analysis of the expression patterns of the *wcs* genes (see Table S3). The 13 tissues were brain, liver, kidney, lung, heart, spleen, muscle, testis, blood, lymph node, skin, placenta, and pancreas. The short reads were mapped on the *wcs* genes with an identity cutoff value of 99%. The *wcs* genes were defined as being expressed if more than one read was mapped (52).

SUPPLEMENTAL MATERIAL

Supplemental material for this article may be found at https://doi.org/10.1128/JVI.00542-19.

SUPPLEMENTAL FILE 1, PDF file, 0.5 MB.

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